MIL-E-1/832A
17 February 1971
SUPERSEDING
MIL-E-1/832(SigC)
5 April 1955

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12AL5

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin diode, very high frequency

Outline	---	5-1 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	1k	2a	h	h	2k	sd	1a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Epp/a	Rl	Cl	epx	Ib/a	Ip/a	i(surge)/a	(A) Ehk	(A) Alt
Unit:	V	Vac	Ohms	μF	v	ma	mAdc	ma	v	ft
Maximum:	13.8	---	---	---	360	60	10	350	360	Note 1
Minimum:	11.4	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	165	11,000	8	---	---	---	---	---	---

GENERAL:

Qualification - Not required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1353	Operation of rectifiers	Ehk = Eo = 117 Vac (see notes 2 and 3)	0.65	II	Io	16	---	mAdc
1231	Emission current	Eb = 10 Vdc (see note 4)	0.65	II	Is	40	---	mAdc
(A) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	138	162	mA
(A) 1336	Heater-cathode leakage	See note 4	---	---	Ihk	---	10	µAdc
1256	Electrode current (anode)	Ebb = 0, Rp = 40,000 ohms (see note 4)	---	---	Ib	2	20	µAdc
---	Electrode current (anode) (differences between sections)		---	---	Ib	---	5	µAdc
1331	Direct-interelectrode capacitance	Shield No. 316	---	---	C1p-C2p	---	0.026	pF
		Shield No. 316	---	---	C1p-h+ 1k+int sd +ext sd	2.3	3.7	pF
		Shield No. 316	---	---	C2p-h+ 2k+int sd +ext sd	2.3	3.7	pF
		Shield No. 316	---	---	C1k-h+ 1p+int sd +ext sd	2.8	4.0	pF
		Shield No. 316	---	---	C2k-h+ 2p+int sd +ext sd	2.8	4.0	pF
(A) 2126	Glass strain		---	---	---	---	---	---
(A) 1121	Base strain		---	---	---	---	---	---
(A) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A: Ehk = -350 Vdc (see note 3)	---	---	---	---	---	---
---	Life-test end point (500 hours)	Operation of rectifiers	---	---	Io	14	---	mAdc
(A)	Periodic-check test							
1031	Low-frequency vibration	No voltages applied	---	---	---	---	---	---

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. In a full-wave circuit, adjust Zp a such that a tube having Etd 10 Vdc at 60 mAdc per anode gives Ia 18 mAdc. The minimum peak current should be 50 ma per anode.
4. Test each unit separately.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU, MI
Air Force - 11, 80
DSA - ES

User activities:
Army - WC, AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2465)

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/832A, Electron Tube, Receiving Type 12AL5

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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